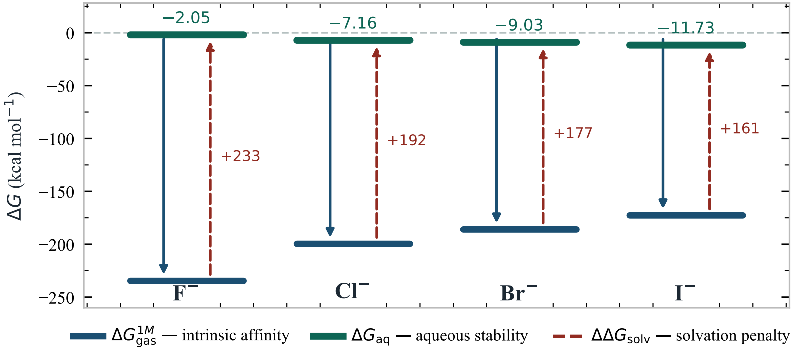
Hydration-Driven Inversion of the Methylmercury Halide Stability Series

CH_3HgI > CH_3HgBr > CH_3HgCl > CH_3HgF (aqueous stability — opposite to gas-phase affinity order)



Halide hydration ($\Delta G_{solv}(X^-)$) overcompensates intrinsic Hg–X affinity, reversing the aqueous stability order.

Fluoride retains only 0.88% · Chloride 3.59% · Bromide 4.85% · Iodide 6.79% of their gas-phase binding energy in solution.